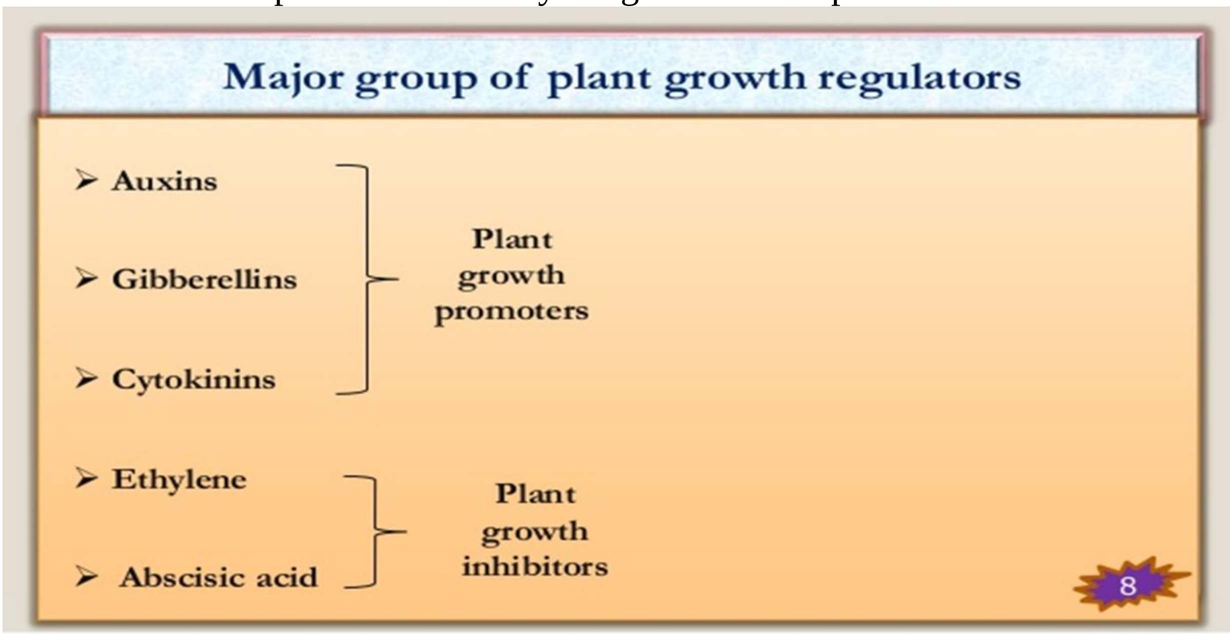
Plant Growth hormones

a) Term Hormones was coined by Starling (1906)
b) Hormones inhibit promote or modify the growth of the plant.



Plant growth promoters

Auxins:

1. The first phytohormone to be discovered is the Auxin and it was discovered by the biologist Charles Darwin.
2. F. Went isolated natural auxin from Avena coleoptile tips.
3. **Precursor: Tryptophan**. Auxins are one of the most important plant hormones.
4. The chief naturally occurring auxin is indole-3 acetic acid – IAA and other related compounds.
5. These plant growth regulators are generally produced at the points of stems and roots from where they are transported to other parts of the plants.
6. These plant hormones include both natural and synthetic sources.
7. Indole-3-acetic acid and are obtained from natural plant sources, whereas indole butyric acid (IBA), naphthalene acetic acid (NAA) and 2, 4-dichlorophenoxyacetic acid are obtained from synthetic sources.

Functions of Auxins

1. Promote flowering in plants
2. Used in the process of plant propagation.
3. Used by gardeners to keep lawns free from weeds.
4. Involved in the initiation of roots in stem cuttings.
5. Prevent dropping of fruits and leaves at early stages.
6. Control xylem differentiation and help in cell division.
7. Auxins are widely used as herbicides to kill dicot weeds.
8. Used for the production of fruit without preceding fertilization.
9. Promote natural detachment (abscission) of older leaves and fruits.
10. Apical dominance may occur in which the growth of lateral buds is inhibited by the growth of apical buds. In such cases, the shoot caps may be removed.
11. These are produced by the apex of root and shoot.

Gibberellins

- Discovered in Japan in rice plants infected by fungus *<u>Gibbereliafujikoroi</u>*(plants become very tall, thin, pale). Kurosawa discovered it.
- Yabuta&Sumiki isolated & extracted crystalline form & named it.
- They are chemically gibberelic acid. 150 types are known GA_1 , GA_2,GA_3….GA_{150}
- They are synthesized on tips of growing root & young leaves.
- They transport in all direction i.e. non polar.
- ***<u>Precursor: Mevalonic acid</u>***

<u>Functions of Gibberellins</u>

- They are cell elongators, increase size of jute fibbers.
- Genetically dwarf plant can be converted into phenotypically tall plant.
- Increase size of fruits
- Seed dormancy can be broken& promote seed germination.
- Seedless fruits grown (parthenocarpy).
- More effective than auxins.

1. Delay senescence in fruits.
2. Involved in leaf expansion.
3. Break bud and seed dormancy.
4. Promote bolting in cabbages and beet.
5. Help fruits like apples to elongate and improve their shape.
6. Used by the brewing industry for increasing the speed on the malting process.
7. Used as the spraying agent to increase sugarcane yield by lengthening of the stem.
8. Used to fasten the maturity period in young conifers and promote early seed production
9. Helps in increasing the crop yield by increasing the height in plants such as sugarcane and increase the axis length in plants such as grape stalks.
10. Gibberellins are acidic in nature.
11. It also delays senescence.

<u>Cytokinin</u>

1. Letham coined the term cytokinin.
2. Stimulate cell division.
3. Skoog& Miller first discovered in Tobacco plant *Nicotianatobacum.*
4. They observed that callus proliferated when nutrient medium was supplemented with coconut milk and degraded sample of DNA (obtained from herring sperm).They named it as kinetin.
5. Kinins are 6-furfuryl amino purine.
6. Natural cytokinin= Zeatin extracted from unripe maize grains.
7. Natural cytokinin arederieved from banana flowers, apple, tomato, coconut milk etc.
8. Synthetic cytokinin= 6-benzyl adenine
9. It is a derivatives of adenine.
10. Synthesized by young plants where active cell division occurs.
11. Coconut milk containcytokinin.
12. **<u>Precursor: Adenine</u>**

<u>Functions of Cytokinins</u>

1. Break bud and seed dormancy.
2. Promotes the growth of the lateral bud.
3. Promotes cell division and apical dominance.
4. They are used to keep flowers fresh for a longer time.
5. Used in tissue culture to induce cell division in mature tissues.
6. Promote lateral shoot growth and adventitious shoot formation.
7. Promote nutrient mobilization which in turn helps delay leaf senescence.
8. Helps in delaying the process of ageing (senescence) in fresh leaf crops like cabbage and lettuce.

9. Involved in the formation of new leaves and chloroplast organelles within the plant cell.
10. Used to induce the development of shoot and roots along with auxin, depending on the ratio.

Plant Growth Inhibitors

Abscisic acid

1. It is a growth inhibitor which was discovered in the 1960s.
2. It was initially called dormant. Later another compound abscisin-II was discovered and are commonly called as abscisic acid.
3. This growth inhibitor is synthesized within the stem, leaves, fruits, and seeds of the plant.
4. Abscisic acid mostly acts as an antagonist to Gibberellic acid.
5. It is also known as the stress hormone as it helps by increasing the tolerance of plants to different kinds of stress.

Functions of Abscisic acid

1. Stimulates closing of stomata in the epidermis.
2. Helps in the development and maturation of seeds.
3. Inhibits plant metabolism and seed germination.
4. It is involved in regulating abscission and dormancy.
5. It is widely used as a spraying agent on trees to regulate dropping of fruits.
6. Induces dormancy in seeds and helps in withstanding desiccation and other unfavourable growth factors.

Ethylene

1. Ethylene is a simple, gaseous plant growth regulator, synthesis by most of the plant organs including ripening fruits and ageing tissues.
2. It is an unsaturated hydrocarbon having double covalent bonds between and adjacent to carbon atoms.
3. Ethylene is used as both plant growth promoters and plant growth inhibitors.
4. Ethylene is synthesized by the ripening fruits and ageing tissues.
5. **Precursor: ethephon**

Functions of Ethylene

1. Ethylene is the most widely used plant growth regulator as it helps in regulating many physiological processes.
2. Induce flowering in the mango tree.
3. Promotes sprouting of potato tubers.
4. Breaks the dormancy of seeds and buds.
5. Enhances respiration rate during ripening of fruits.
6. Applied to rubber trees to stimulate the flow of latex.
7. Promotes abscission and senescence of both leaves and flowers.
8. Used to stimulate the ripening of fruits. For example, tomatoes and citrus fruits.
9. Affects horizontal growth of seedlings and swelling of the axis in dicot seedlings.

10. Increases root growth and root hair formation, therefore helping plants to increase their absorption surface area.

Table 31.1

Table 31.1 Overview of Plant Hormones

Hormone	Where Produced or Found in Plant	Major Functions
Auxin (IAA)	Shoot apical meristems and young leaves are the primary sites of auxin synthesis. Root apical meristems also produce auxin, although the root depends on the shoot for much of its auxin. Developing seeds and fruits contain high levels of auxin, but it is unclear whether it is newly synthesized or transported from maternal tissues.	Stimulates stem elongation (low concentration only); promotes the formation of lateral and adventitious roots; regulates development of fruit; enhances apical dominance; functions in phototropism and gravitropism; promotes vascular differentiation; retards leaf abscission.
Cytokinins	These are synthesized primarily in roots and transported to other organs, although there are many minor sites of production as well.	Regulate cell division in shoots and roots; modify apical dominance and promote lateral bud growth; promote movement of nutrients into sink tissues; stimulate seed germination; delay leaf senescence.
Gibberellins	Meristems of apical buds and roots, young leaves, and developing seeds are the primary sites of production.	Stimulate stem elongation, pollen development, pollen tube growth, fruit growth, and seed development and germination; regulate sex determination and the transition from juvenile to adult phases.
Brassinosteroids	These compounds are present in all plant tissues, although different intermediates predominate in different organs. Internally produced brassinosteroids act near the site of synthesis.	Promote cell expansion and cell division in shoots; promote root growth at low concentrations; inhibit root growth at high concentrations; promote xylem differentiation and inhibit phloem differentiation; promote seed germination and pollen tube elongation.
Abscisic acid (ABA)	Almost all plant cells have the ability to synthesize abscisic acid, and its presence has been detected in every major organ and living tissue; may be transported in the phloem or xylem.	Inhibits growth; promotes stomatal closure during drought stress; promotes seed dormancy and inhibits early germination; promotes leaf senescence; promotes desiccation tolerance.
Ethylene	This gaseous hormone can be produced by most parts of the plant. It is produced in high concentrations during senescence, leaf abscission, and the ripening of some types of fruit. Synthesis is also stimulated by wounding and stress.	Promotes ripening of many types of fruit, leaf abscission, and the triple response in seedlings (inhibition of stem elongation, promotion of lateral expansion, and horizontal growth); enhances the rate of senescence; promotes root and root hair formation; promotes flowering in the pineapple family.

Assignment:

1. Which hormone is employed for artificial ripening of banana fruits?
2. Enlist the plant growth promoters.
3. Enlist the plant growth inhibitors
4. Enlist the plant growth regulators.
5. Give the full form of IAA, IBA, NAA.
6. Which hormone is used to induce parthenocarpy.
7. Which hormone helps in fruit ripening
8. Give precursors of auxin
9. Give precursors ofcytokinin
10. Give precursors of ethylene
11. Which hormone is called stress hormones.
12. Which hormone acts as anti ageingagent.